Biochemistry



Section N
Nucleotide Metabolism

Guodong Yang

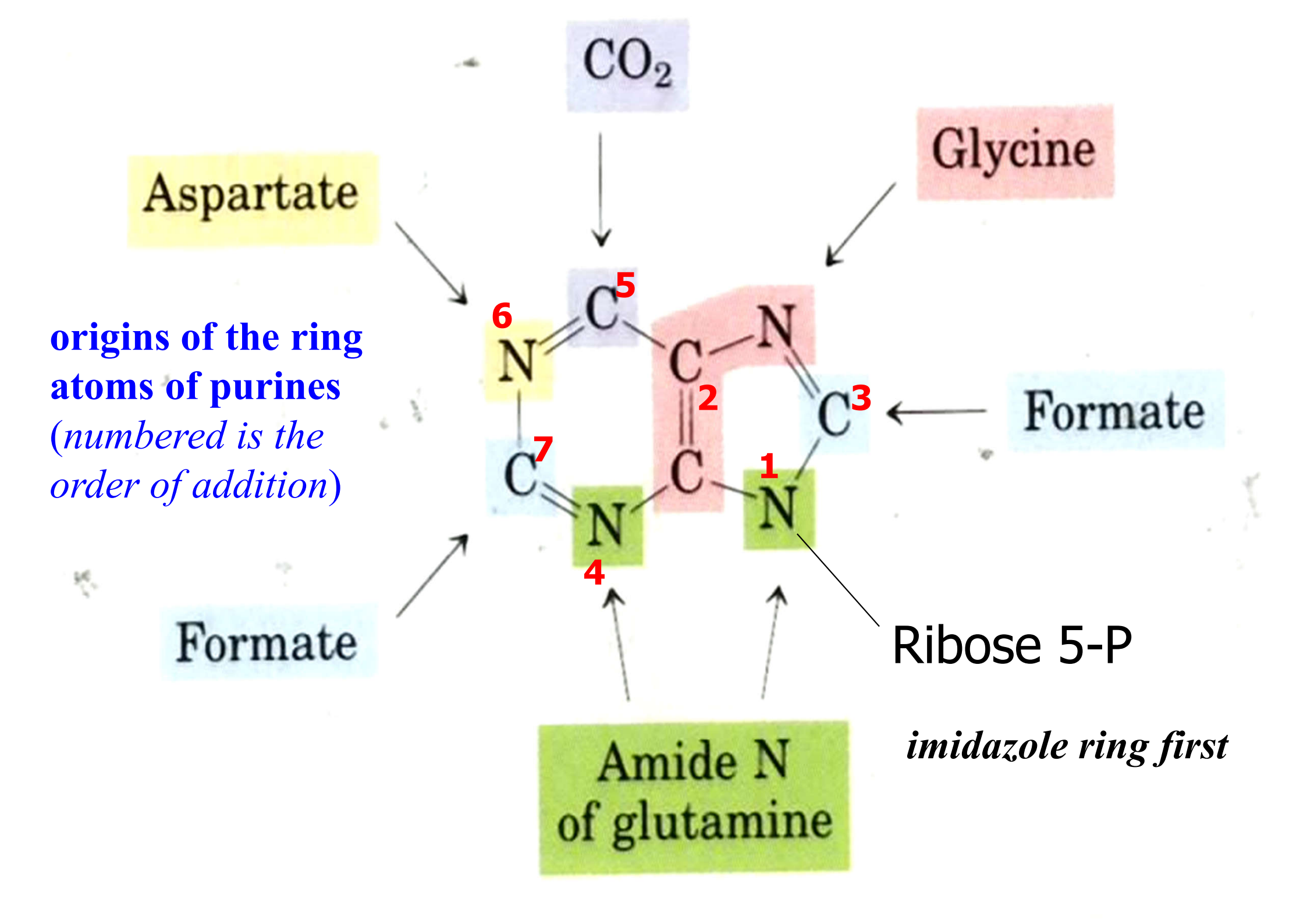
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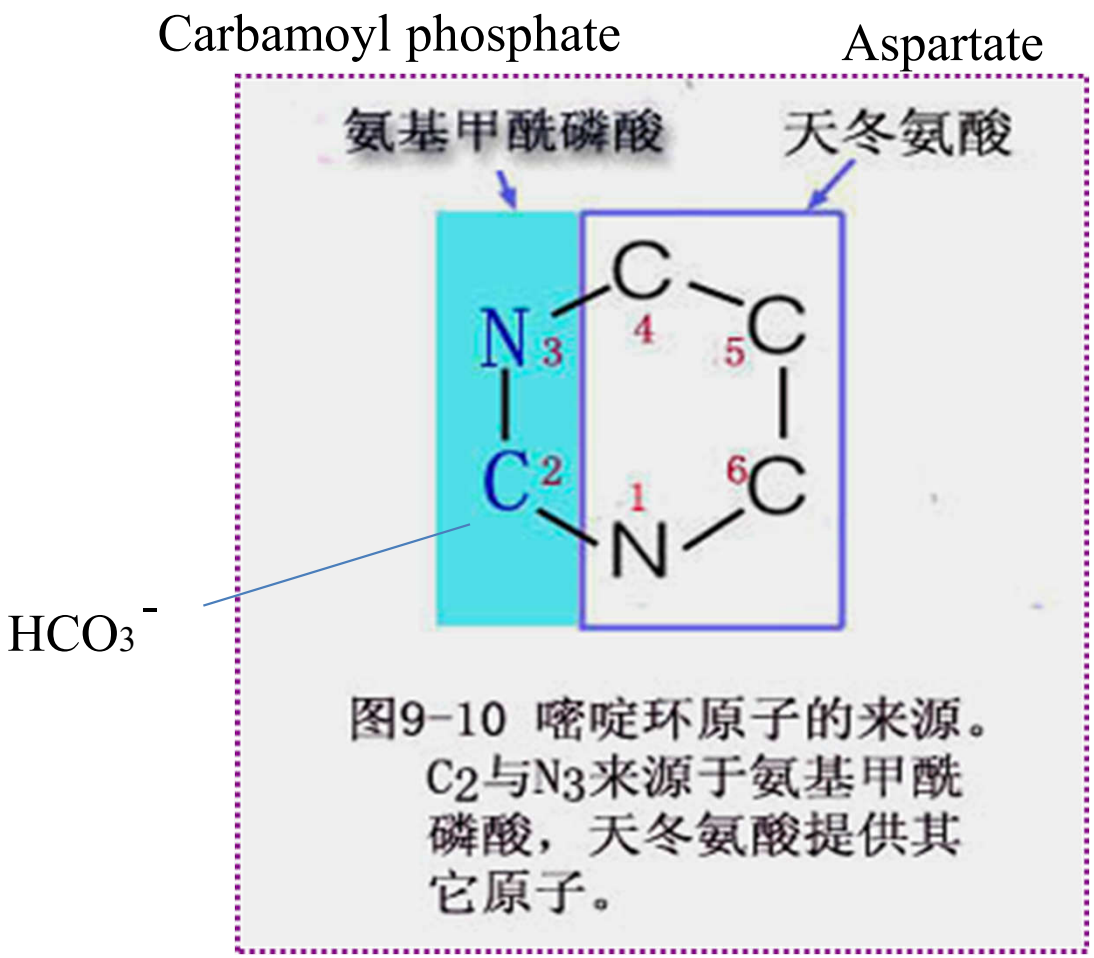
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13854808762

Radioisotope tracer experiments revealed the origins of the atoms in the purine and pyrimidine rings




CO_2
Glycine
Aspartate
origins of the ring atoms of purines (*numbered is the order of addition*)
Formate
Formate
Ribose 5-P
Amide N of glutamine
imidazole ring first

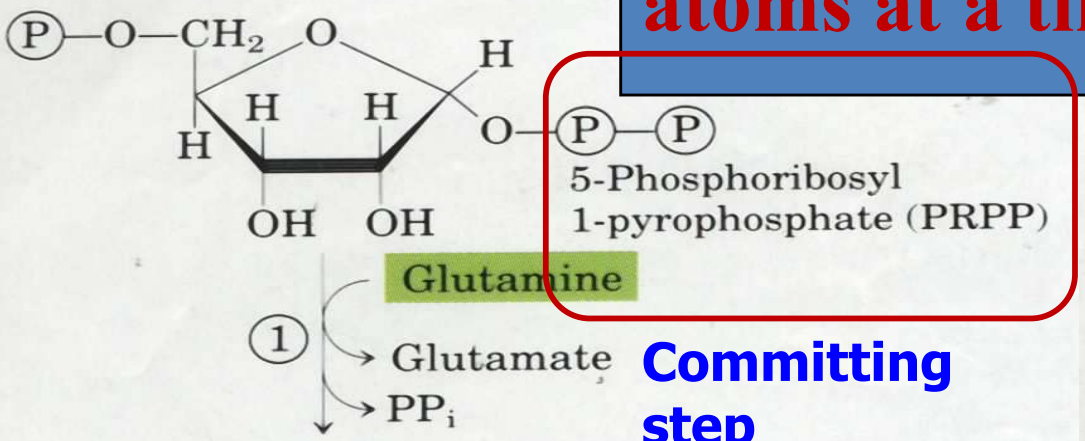


The atoms of the pyrimidine rings were revealed to be derived from HCO_3^-, Gln and Asp.

***De novo* purine nucleotide synthesis: the base assembles on the ribose phosphate; IMP is first nucleotide synthesized.**



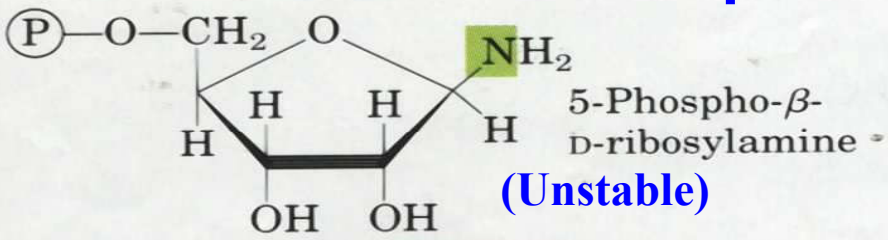
The purine ring is built up one or three atoms at a time on ribose 5-P



Glutamine

① → Glutamate + PP_i

Committing step

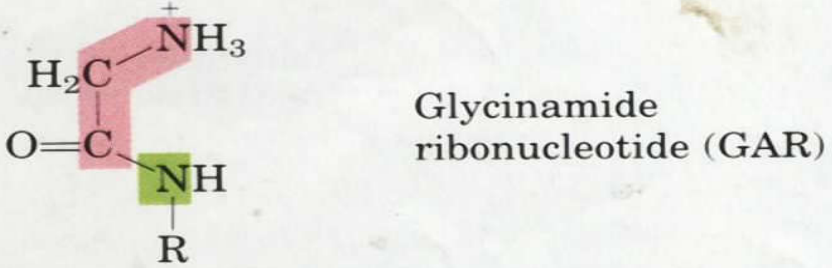


(Unstable)

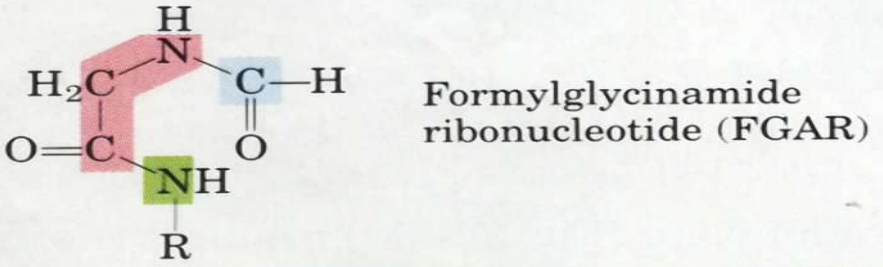
Glycine

② ATP → ADP + P_i

1, 3, 5 are catalyzed by one protein!

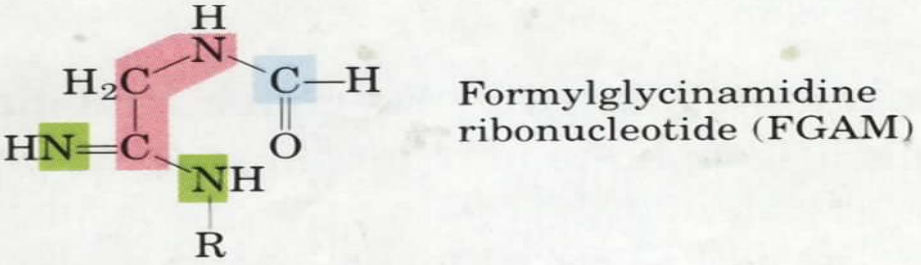


③ N^{10}-Formyl H_4 folate → H_4 folate

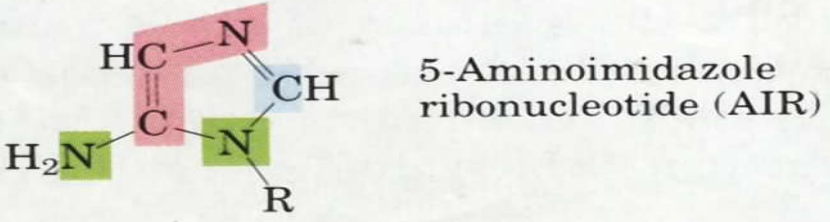


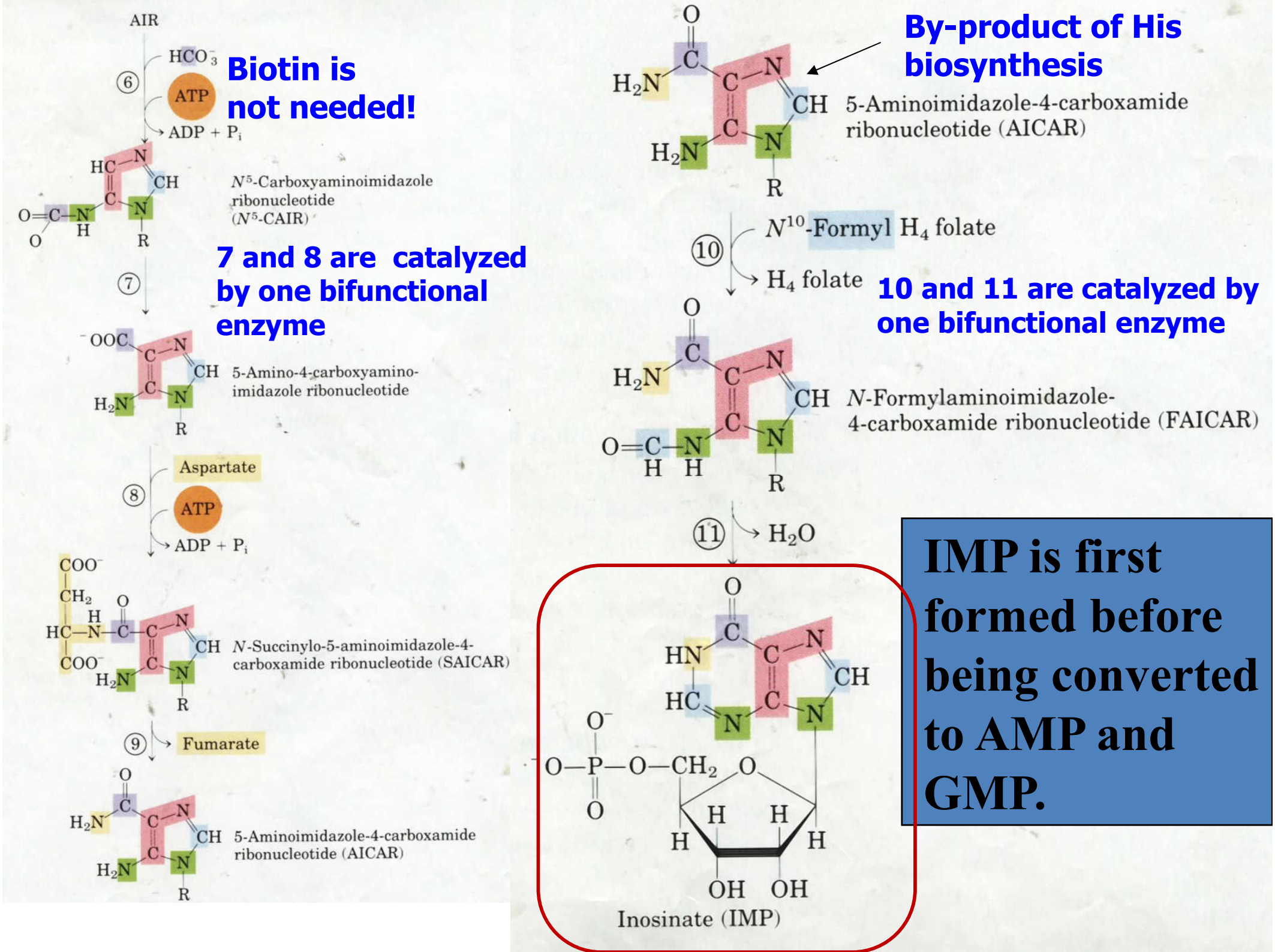
Glutamine

④ → Glutamate + ATP → ADP + P_i



⑤ ATP → ADP + P_i + H_2O





Biotin is not needed!

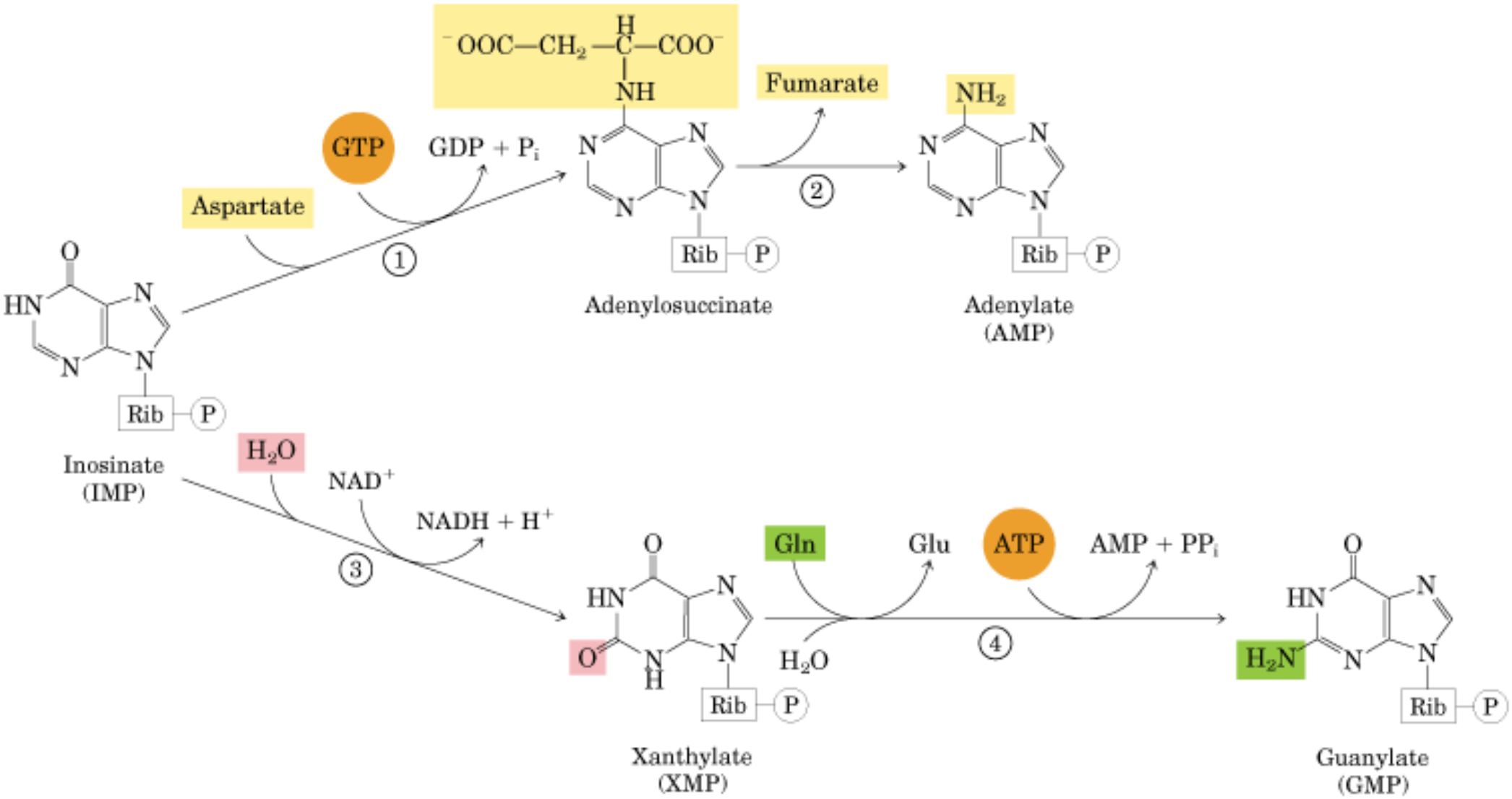
7 and 8 are catalyzed by one bifunctional enzyme

10 and 11 are catalyzed by one bifunctional enzyme

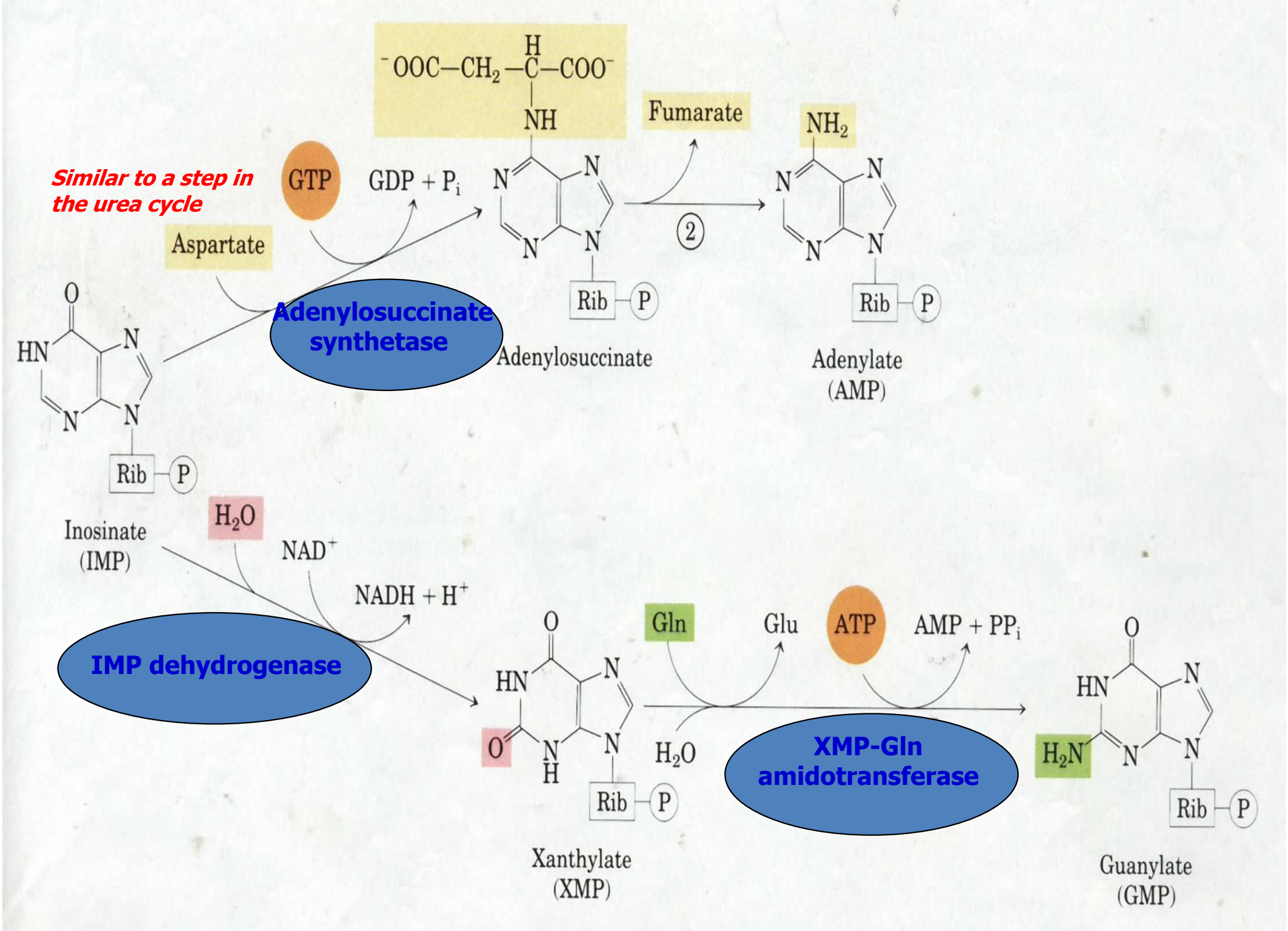
IMP is first formed before being converted to AMP and GMP.

De novo purine nucleotide synthesis: AMP and GMP are derived from IMP.



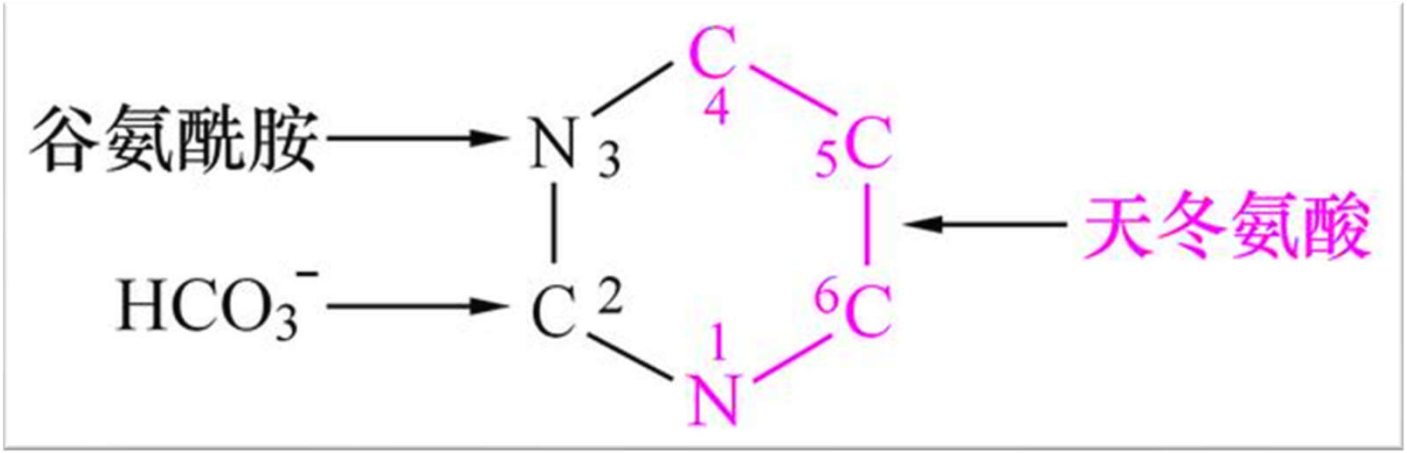

⁻OOC—CH₂—C(H)(NH)—COO⁻
Aspartate
GTP
GDP + Pi
Fumarate
NH2
Rib
P
Adenylosuccinate
Adenylate
(AMP)
Inosinate
(IMP)
H2O
NAD+
NADH + H+
Gln
Glu
ATP
AMP + PPi
Xanthylate
(XMP)
Guanylate
(GMP)

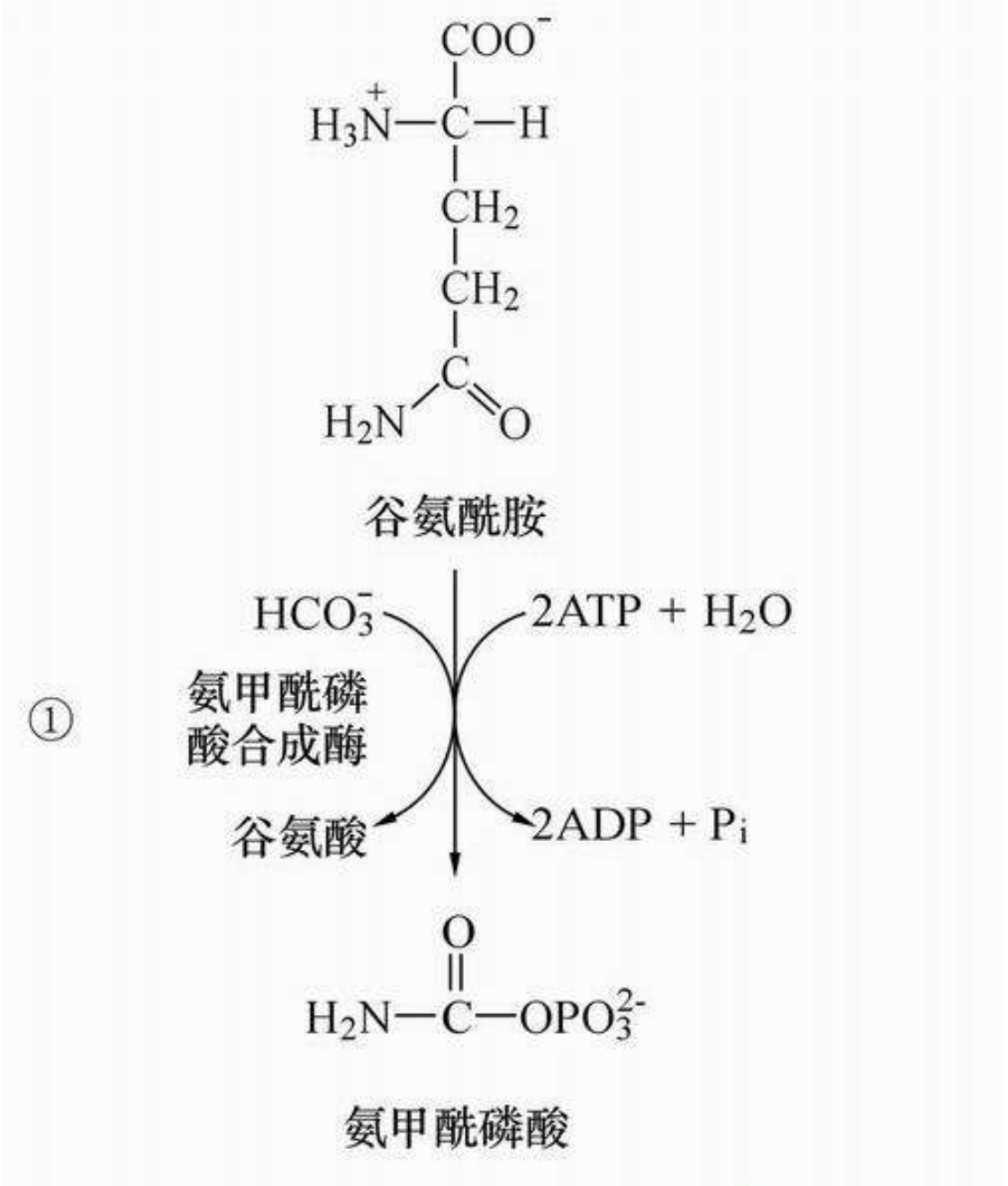
Next

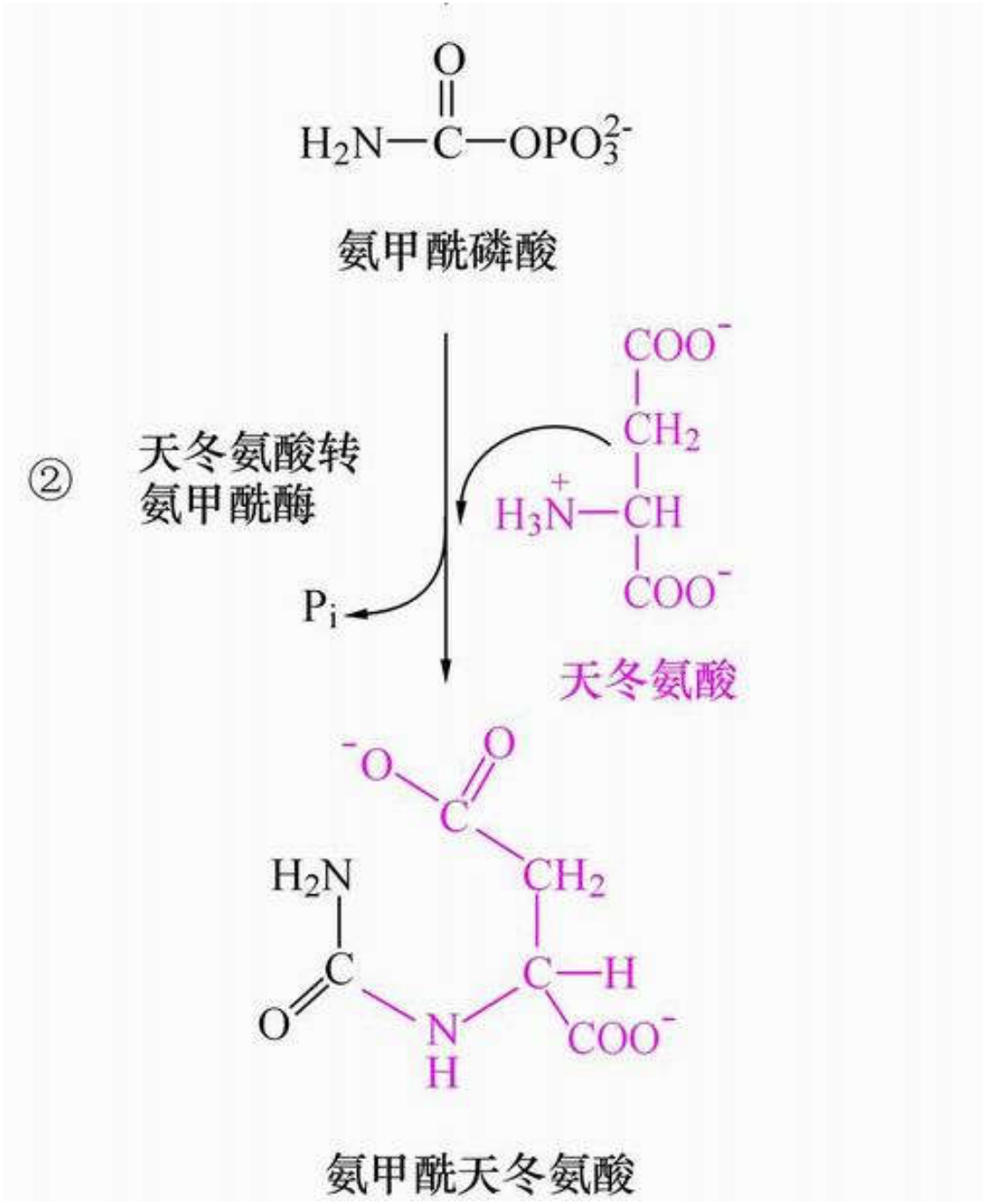

Similar to a step in the urea cycle
GTP
GDP + P_i
Aspartate
Adenylosuccinate synthetase
Fumarate
2
Rib
P
Adenylosuccinate
Adenylate (AMP)
Inosinate (IMP)
H_2O
NAD^+
$NADH + H^+$
IMP dehydrogenase
Gln
Glu
ATP
$AMP + PP_i$
H_2O
XMP-Gln amidotransferase
Xanthylate (XMP)
Guanylate (GMP)

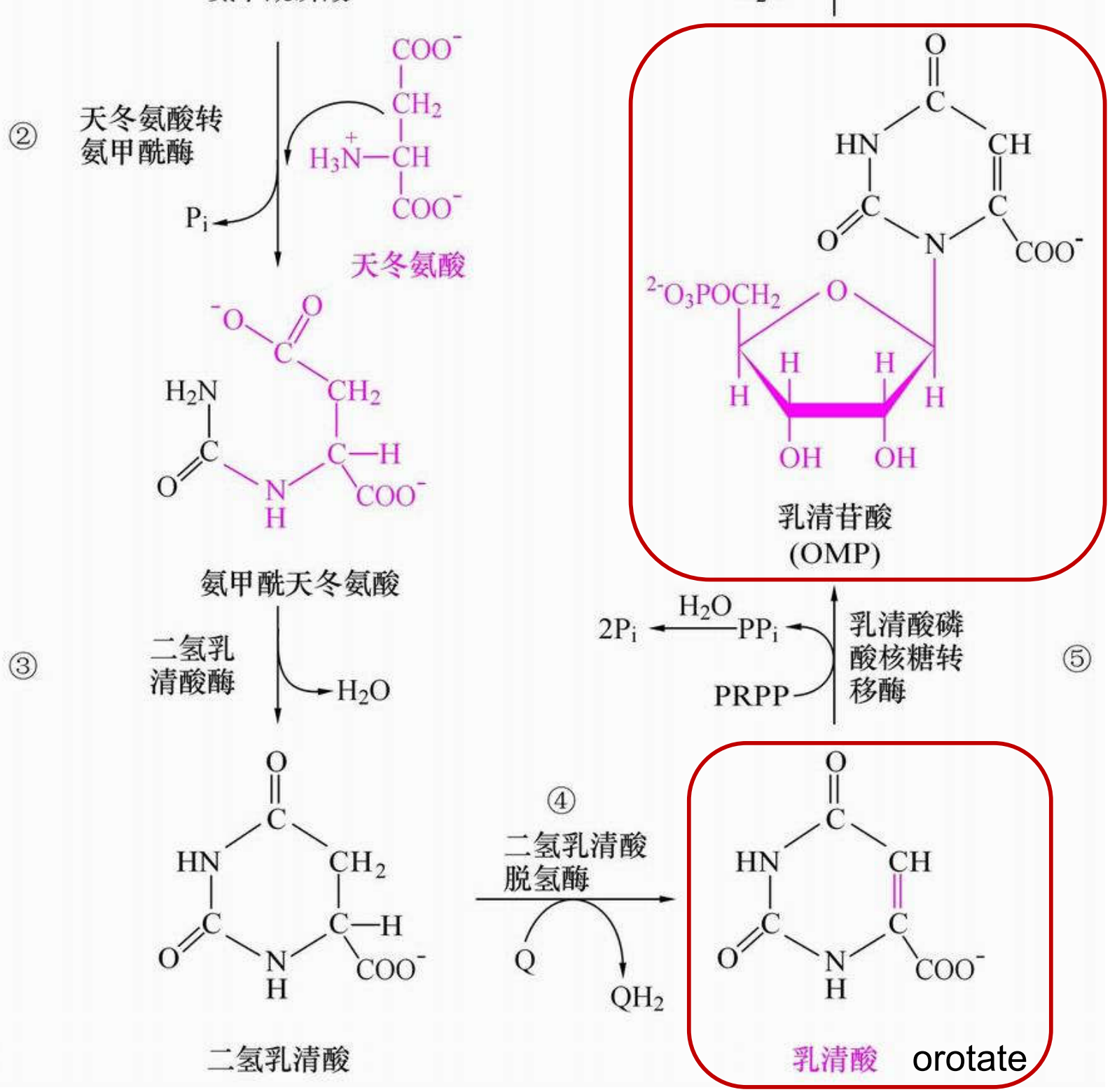
The *de novo* biosynthesis of pyrimidine nucleotides: the base ring (as orotate) is first assembled before being attached to the phosphoribosyl group to form orotidylate

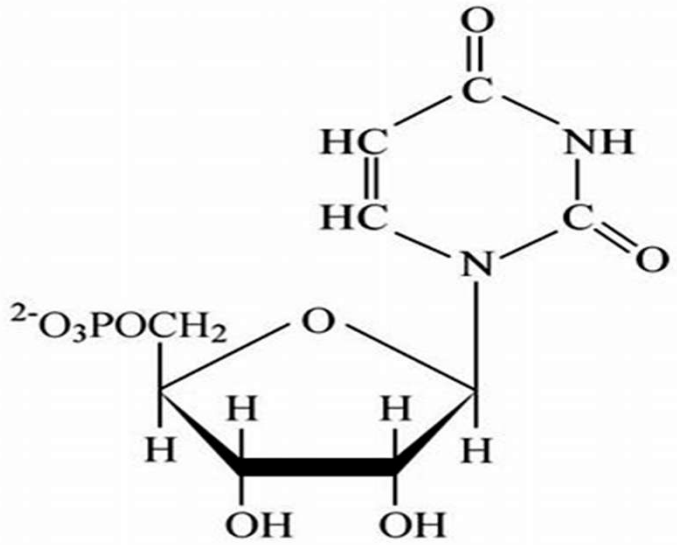



谷氨酰胺
HCO_3^-
N 3
C 2
C 4
5C
6C
1 N
天冬氨酸


COO⁻
H₃N⁺—C—H
CH₂
CH₂
C
H₂N
O
谷氨酰胺
①
HCO₃⁻
2ATP + H₂O
氨甲酰磷
酸合成酶
谷氨酸
2ADP + Pi
O
H₂N—C—OPO₃²⁻
氨甲酰磷酸


氨甲酰磷酸
②
天冬氨酸转
氨甲酰酶
Pi
天冬氨酸
氨甲酰天冬氨酸

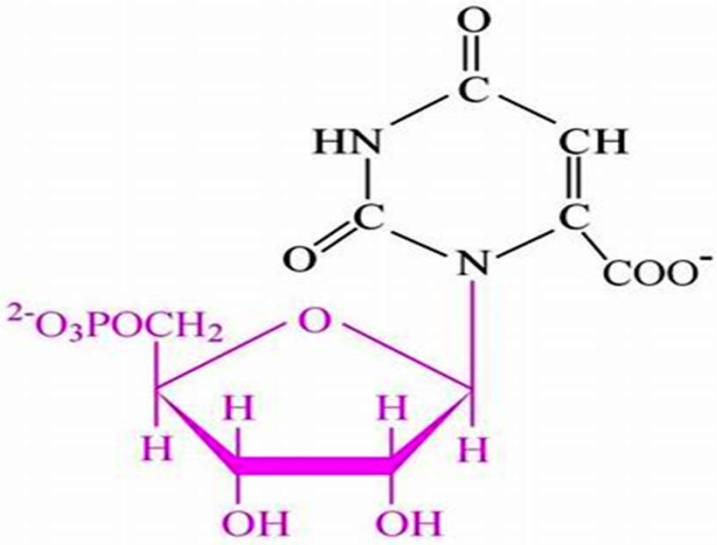

②
天冬氨酸转
氨甲酰酶
Pi
天冬氨酸
氨甲酰天冬氨酸
③
二氢乳
清酸酶
H2O
二氢乳清酸
④
二氢乳清酸
脱氢酶
Q
QH2
乳清酸 orotate
⑤
乳清酸磷
酸核糖转
移酶
2Pi
H2O
PPi
PRPP
乳清苷酸
(OMP)



尿苷酸(UMP)

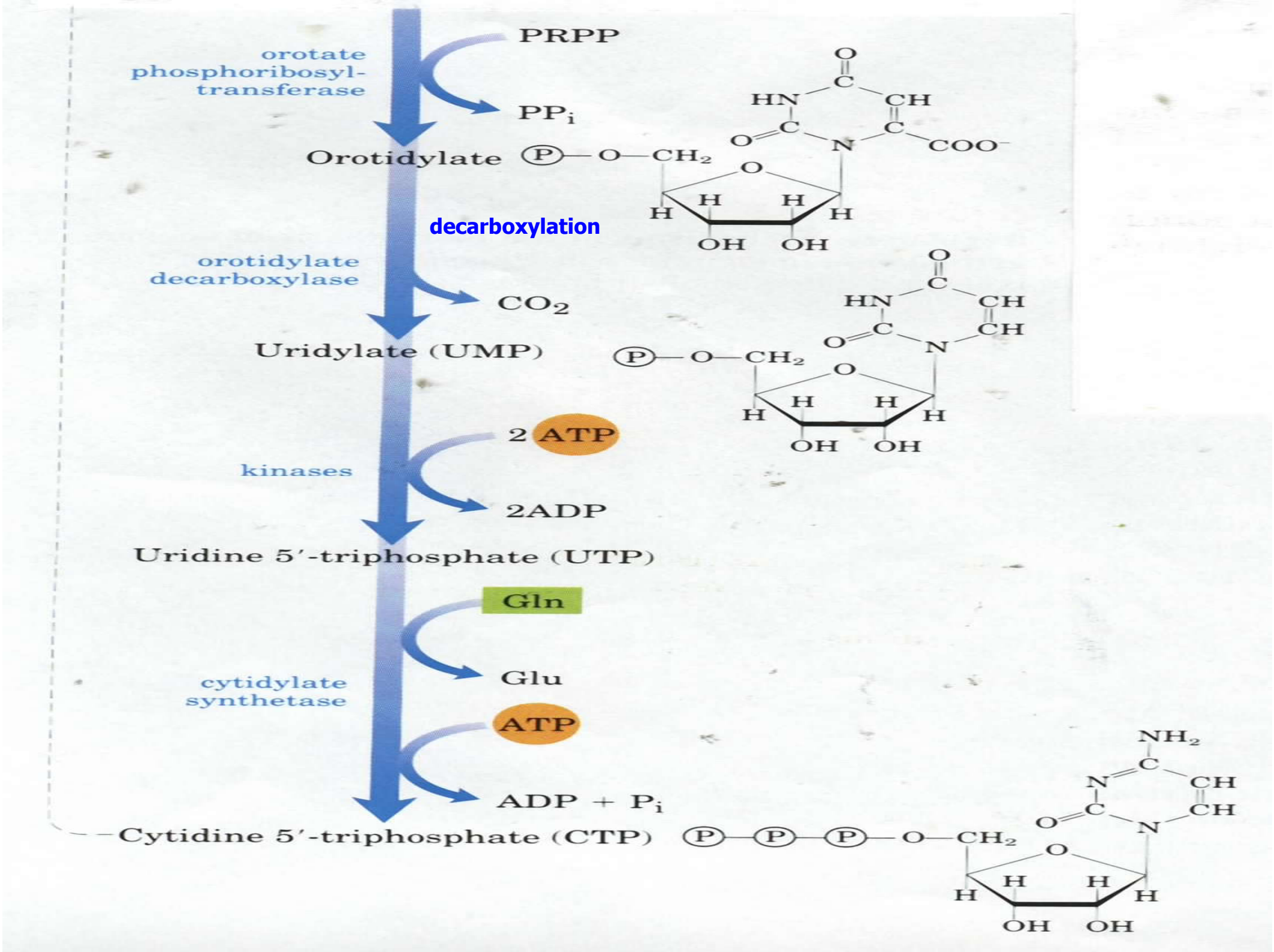


⑥



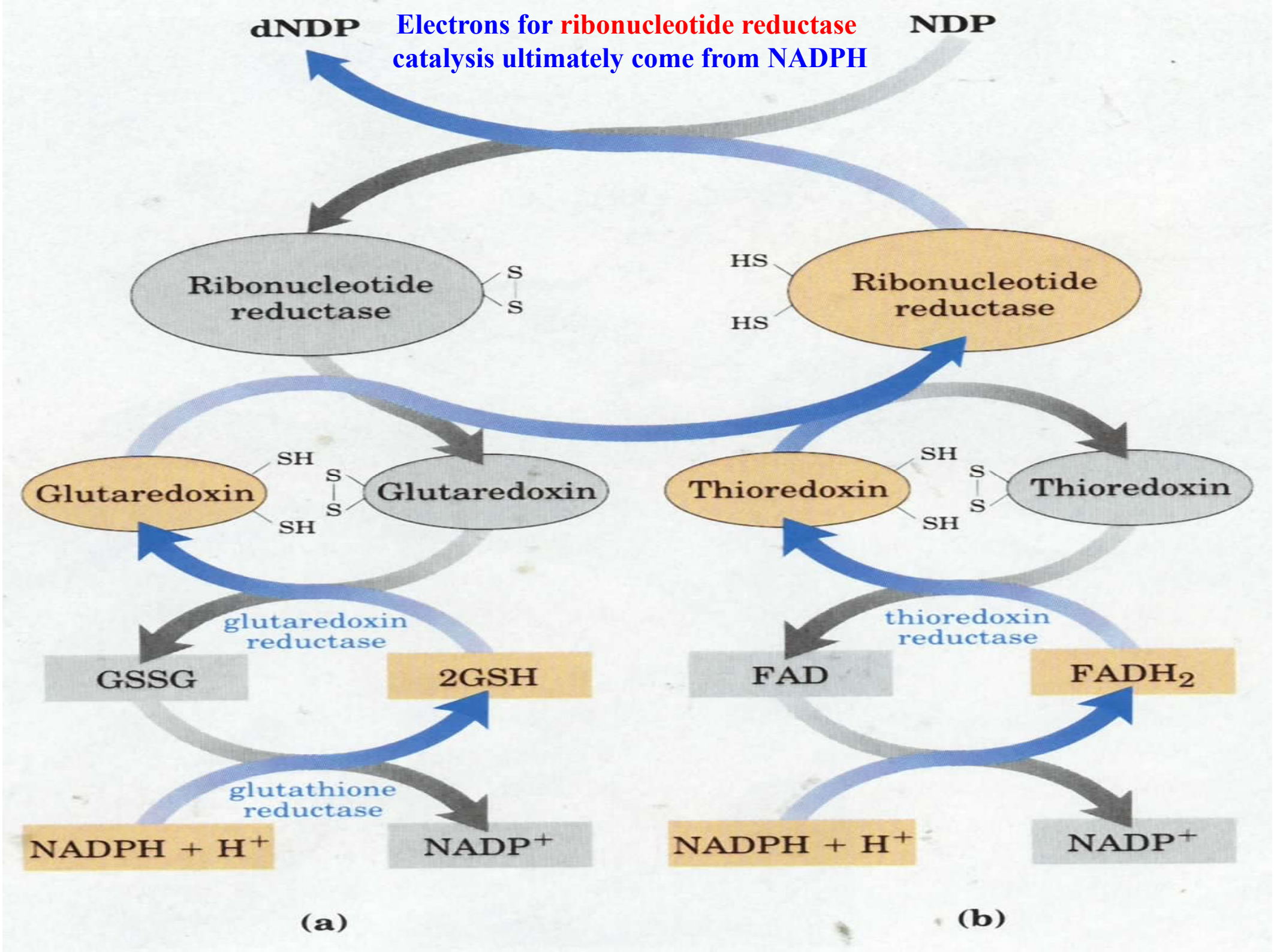
乳清苷酸
(OMP)

UTP and CTP are then derived from orotidylate *via* decarboxylation and amination reactions


PRPP
orotate phosphoribosyl-transferase
PP_i
Orotidylate
decarboxylation
orotidylate decarboxylase
CO_2
Uridylate (UMP)
2 ATP
kinases
2ADP
Uridine 5′-triphosphate (UTP)
Gln
Glu
cytidylate synthetase
ATP
ADP + P_i
Cytidine 5′-triphosphate (CTP)

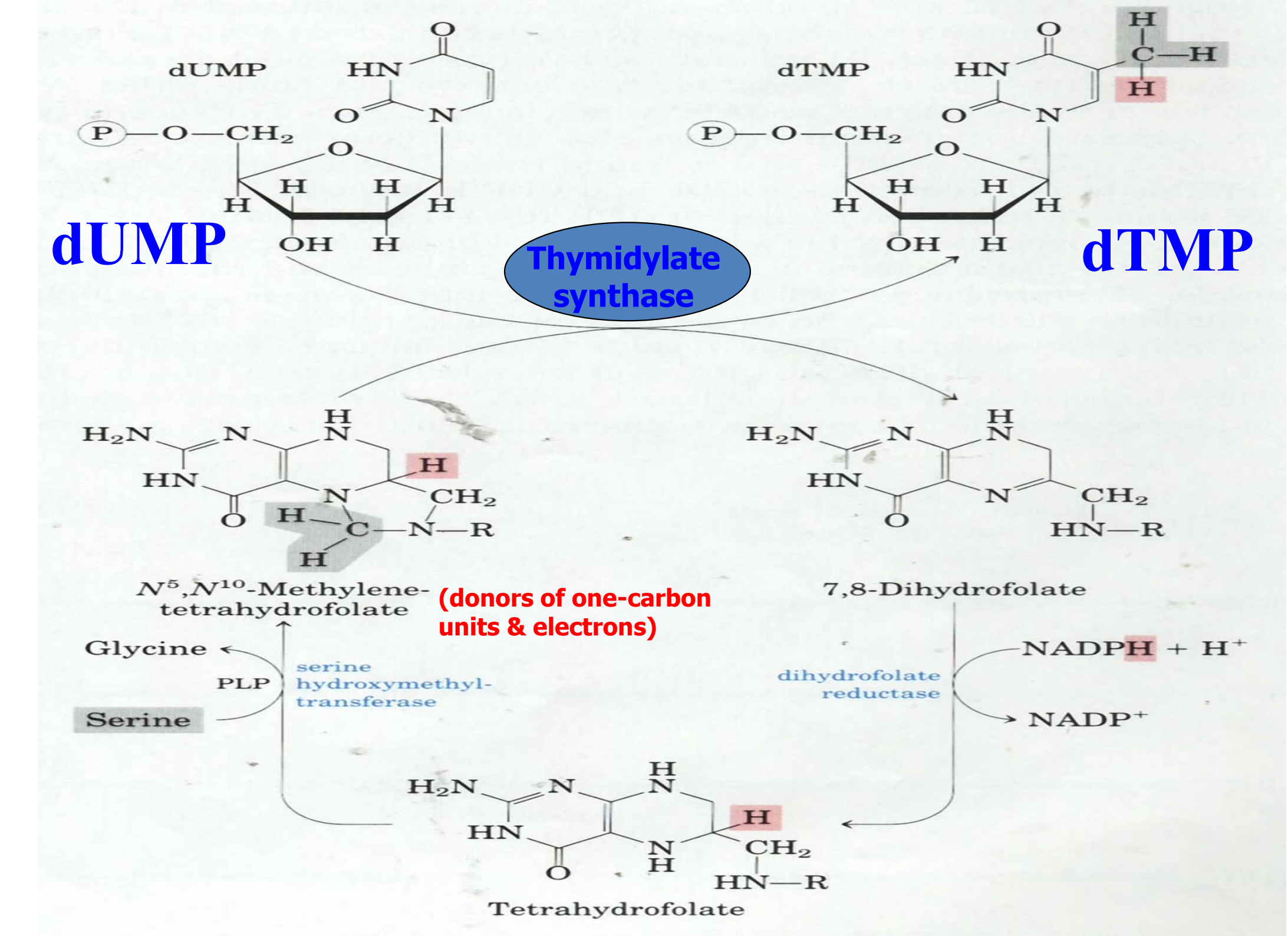
All dNDPs are derived from NDPs via the catalysis of a common ribonucleotide reductase.

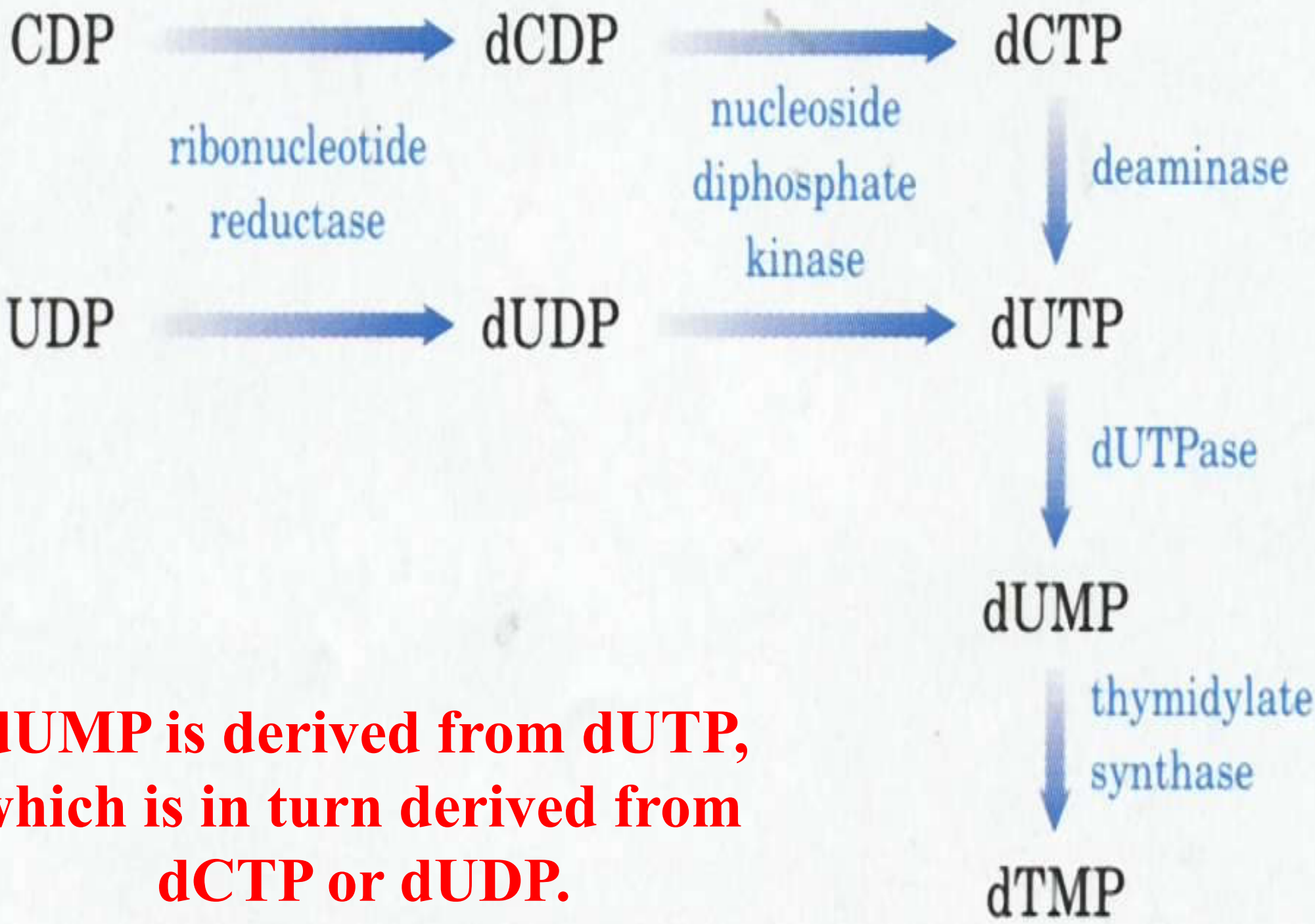



dNDP
Electrons for ribonucleotide reductase catalysis ultimately come from NADPH
NDP
Ribonucleotide reductase
S
S
HS
HS
Ribonucleotide reductase
Glutaredoxin
SH
SH
S
S
Glutaredoxin
Thioredoxin
SH
SH
S
S
Thioredoxin
glutaredoxin reductase
thioredoxin reductase
GSSG
2GSH
FAD
FADH2
glutathione reductase
NADPH + H+
NADP+
NADPH + H+
NADP+
(a)
(b)

dTMP is synthesized by the methylation of dUMP

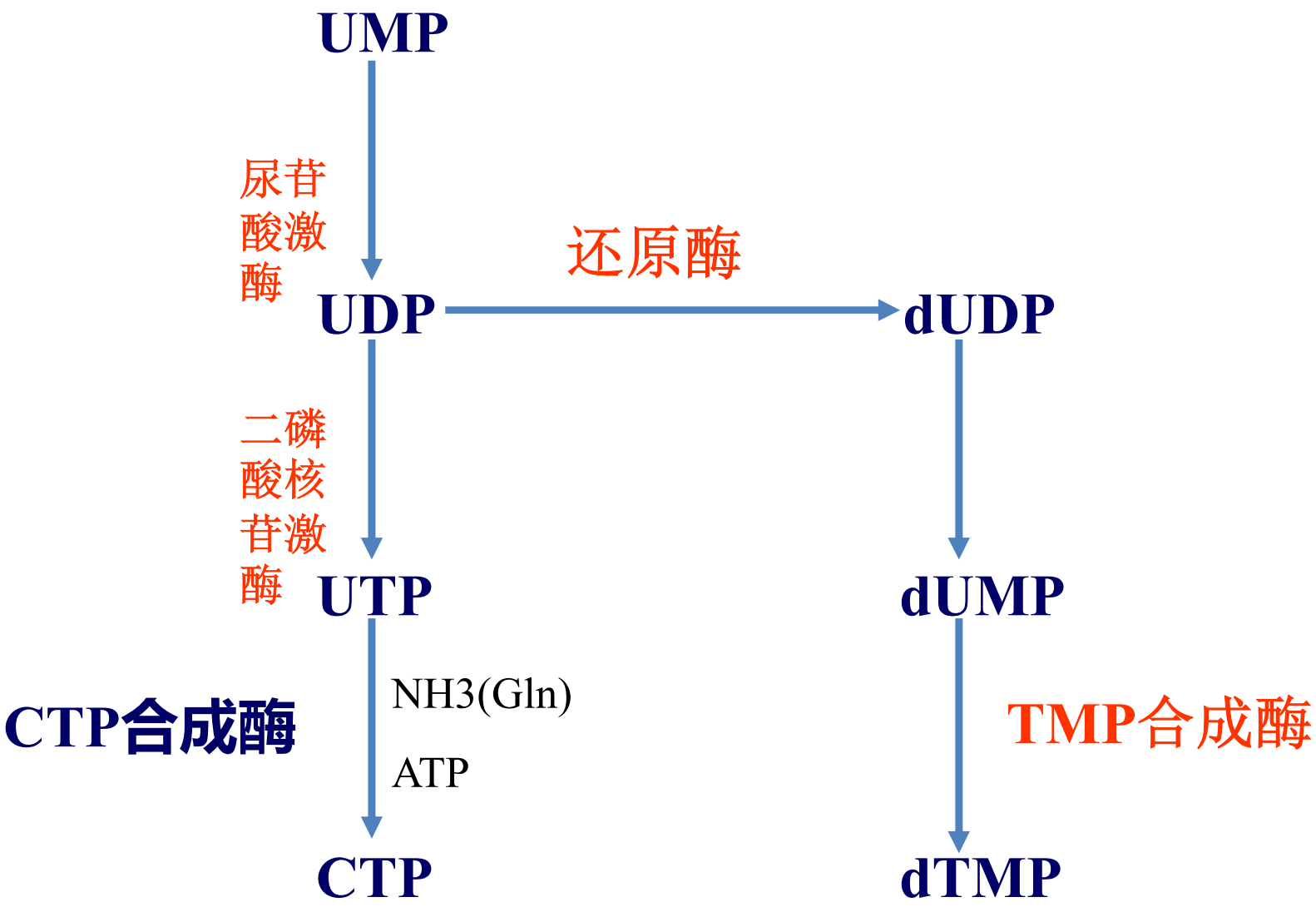



dUMP
dTMP
dUMP
Thymidylate synthase
dTMP
N^5,N^{10}-Methylene-tetrahydrofolate
(donors of one-carbon units & electrons)
7,8-Dihydrofolate
Glycine
PLP
Serine
serine hydroxymethyl-transferase
dihydrofolate reductase
NADPH + H+
NADP+
Tetrahydrofolate



dUMP is derived from dUTP, which is in turn derived from dCTP or dUDP.

UMP转变为CTP和dTMP



Biosynthesis Nucleotide

IMP (Inosinate肌苷酸、次黄嘌呤核苷酸) ------**AMP**, **GMP**

O (orotate乳清酸）------OMP(乳清苷酸)-------**UMP**----UTP------**CTP**

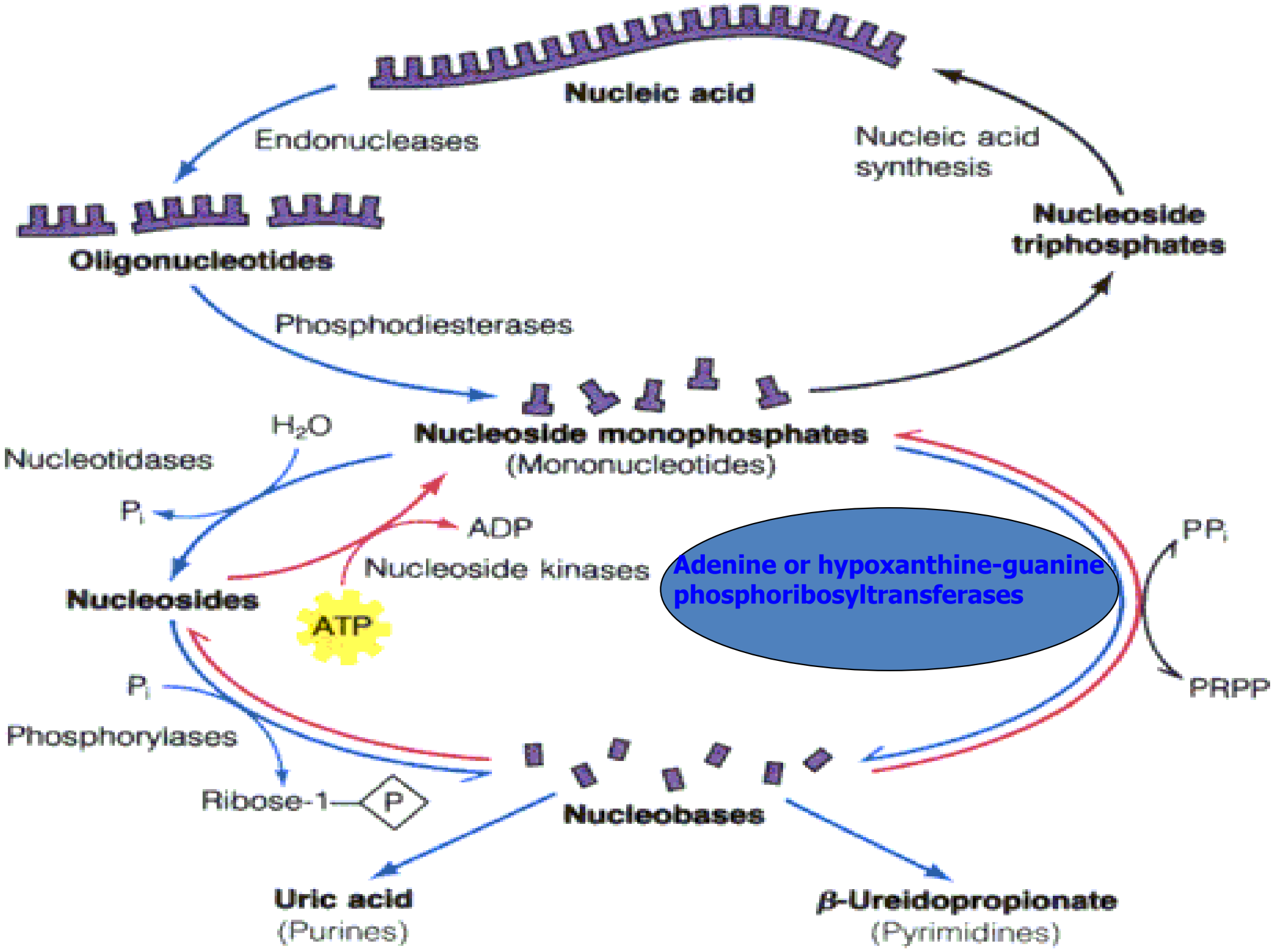
Biosynthesis Deoxynucleotide

ADP-----**dADP**
GDP----**dGDP**
CDP-----**dCDP**

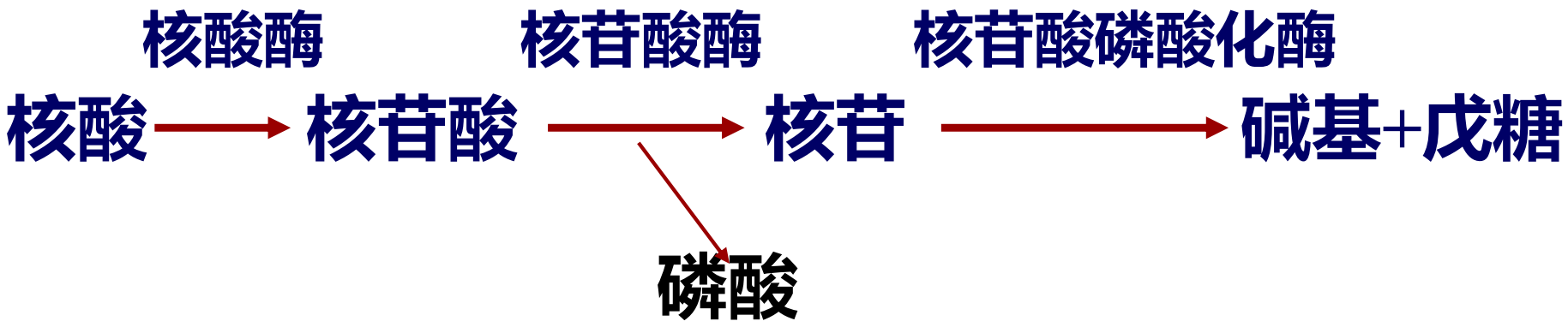
UDP----dUDP----dUMP----**dTMP**

Purine and pyrimidine bases can be reconverted to nucleotides via the salvage pathway




Nucleic acid
Endonucleases
Nucleic acid synthesis
Oligonucleotides
Nucleoside triphosphates
Phosphodiesterases
Nucleoside monophosphates
(Mononucleotides)
H_2O
Nucleotidases
P_i
ADP
Nucleoside kinases
Nucleosides
ATP
Adenine or hypoxanthine-guanine phosphoribosyltransferases
PP_i
PRPP
P_i
Phosphorylases
Ribose-1—P
Nucleobases
Uric acid
(Purines)
β-Ureidopropionate
(Pyrimidines)

核酸的酶促降解



Next

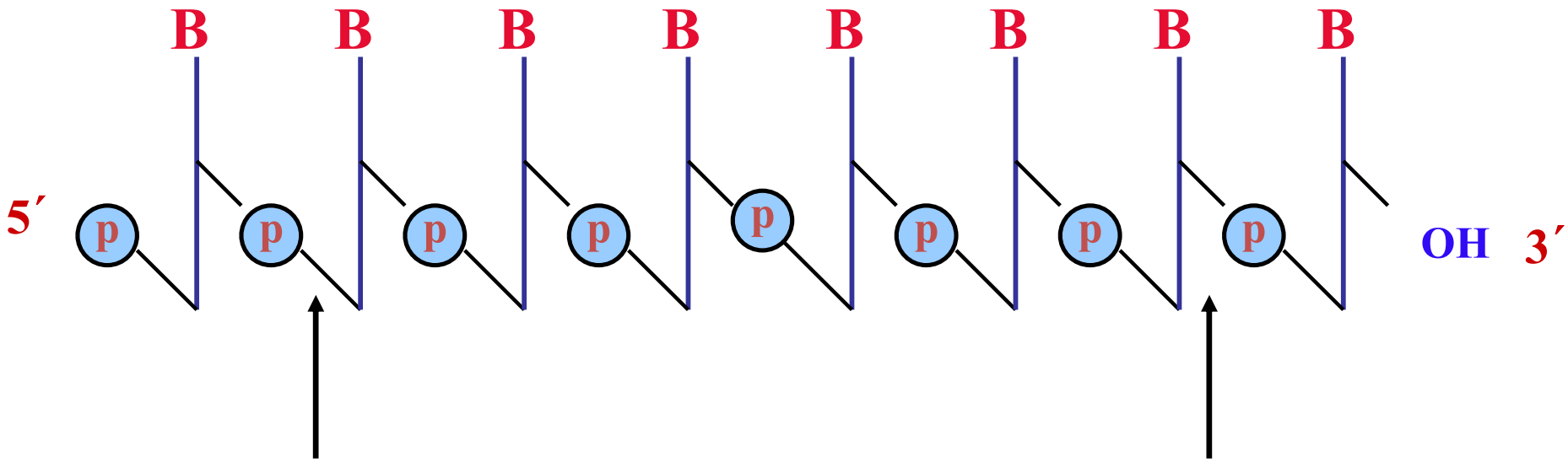
核　酸　酶

1、核酸酶的分类

（1）根据对底物的专一性分为
- 核糖核酸酶(RNase)
- 脱氧核糖核酸酶(DNase)
- 非特异性核酸酶

（2）根据切割位点分为
- 核酸内切酶
- 核酸外切酶
- 核酸内切兼外切酶

1、外切核酸酶对核酸的水解位点

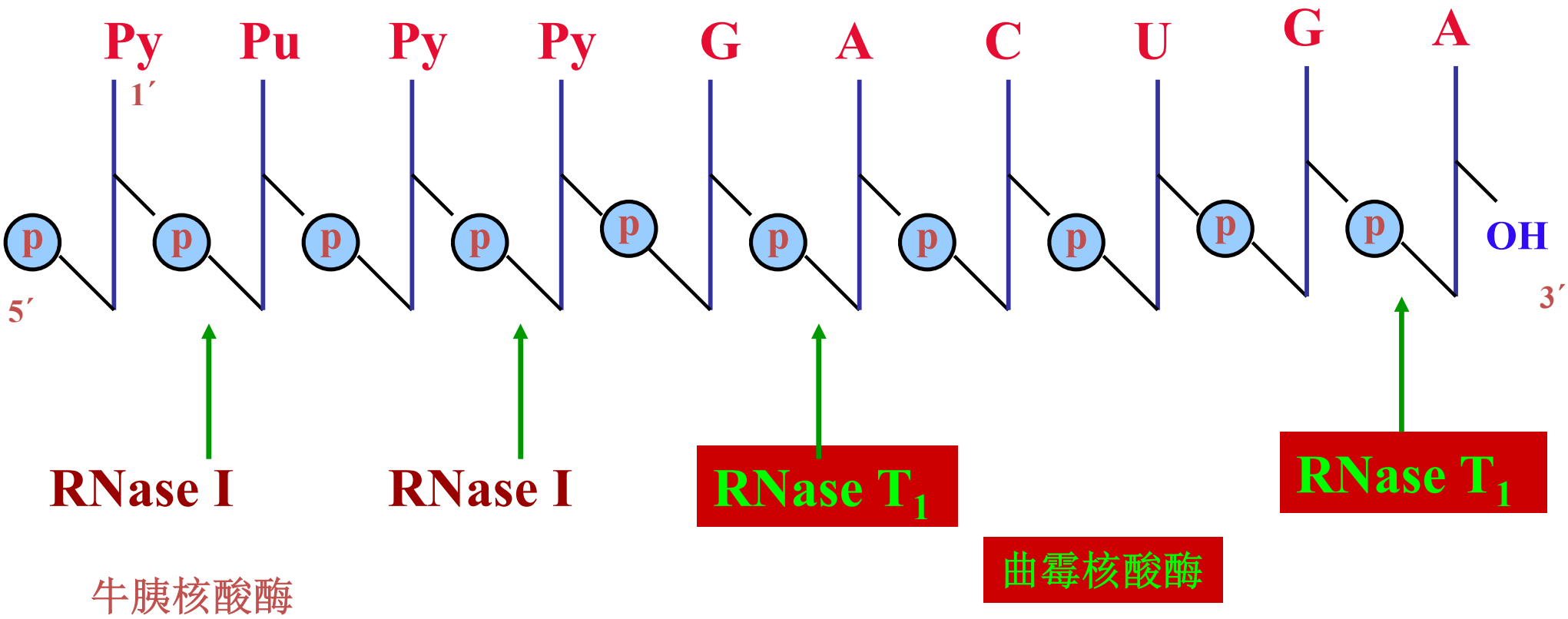


牛脾磷酸二酯酶
（5′端外切5 → 3）

蛇毒磷酸二酯酶
（3′端外切3 → 5）

非特异性核酸酶，作用于DNA、RNA

2、内切核酸酶对RNA的水解位点示意图

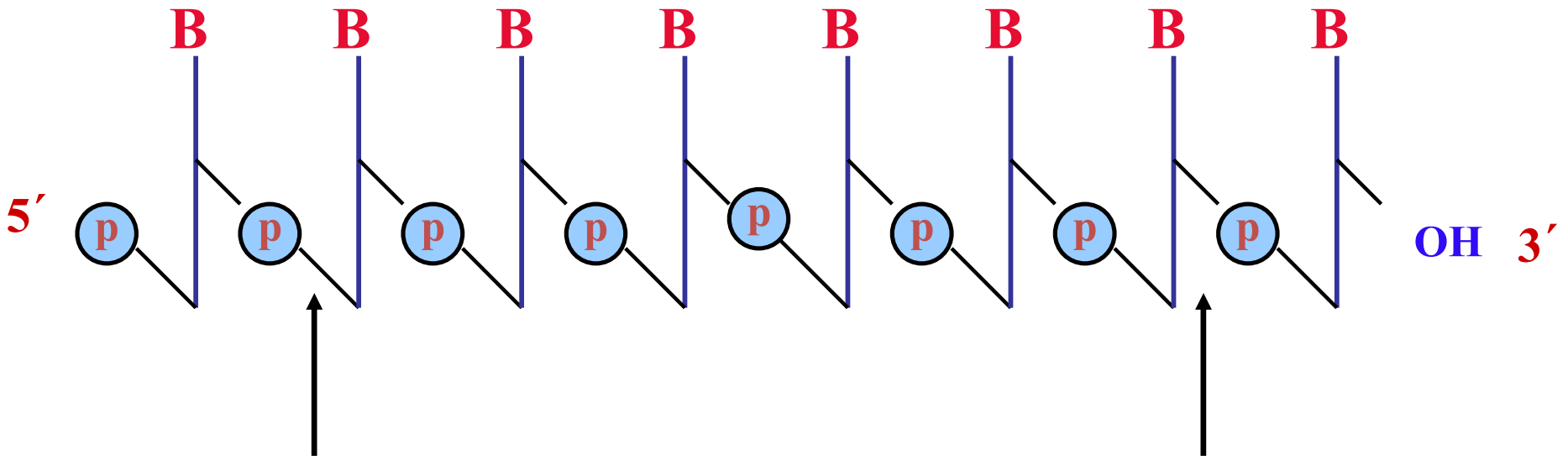


Pu： 嘌呤　　Py： 嘧啶

特异性核酸酶，只作用于RNA或DNA

3、脱氧核糖核酸酶

特异性核酸酶，只作用于DNA

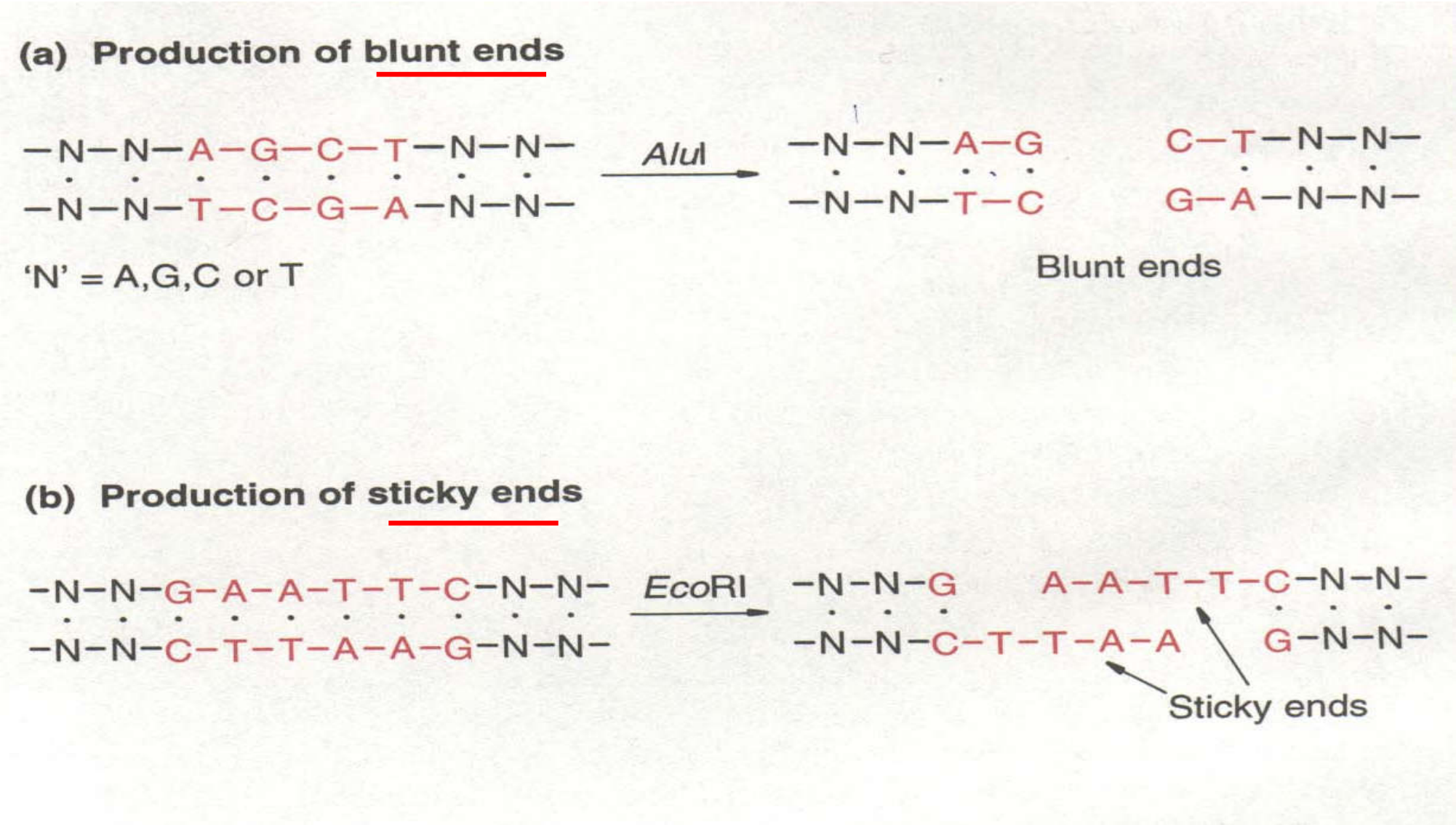


限制性内切酶

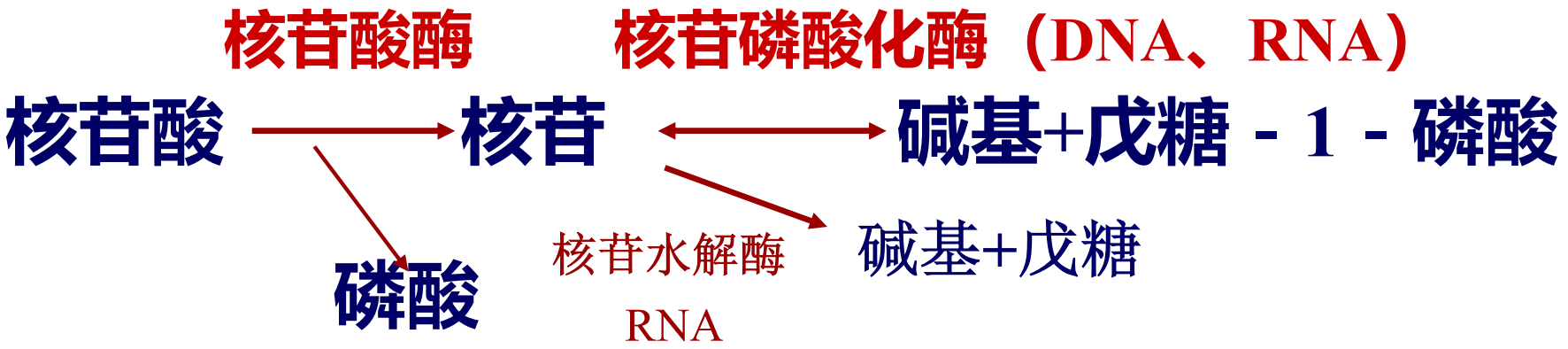
原核生物中存在着一类能识别外源DNA双螺旋中4-8个碱基对所组成的特异的序列，并在此序列的某位点水解DNA双螺旋链，产生粘性末端或平齐末端，这类酶称为限制性内切酶（restriction endonuclease）。

具有很强的专一性， 有特异的识别位点，通常具有二重旋转对称性的回文序列 (palindromic sequence)




(a) Production of blunt ends
–N–N–A–G–C–T–N–N–
–N–N–T–C–G–A–N–N–
AluI
–N–N–A–G
–N–N–T–C
C–T–N–N–
G–A–N–N–
'N' = A,G,C or T
Blunt ends
(b) Production of sticky ends
–N–N–G–A–A–T–T–C–N–N–
–N–N–C–T–T–A–A–G–N–N–
EcoRI
–N–N–G
–N–N–C–T–T–A–A
A–A–T–T–C–N–N–
G–N–N–
Sticky ends

核苷酸分解代谢



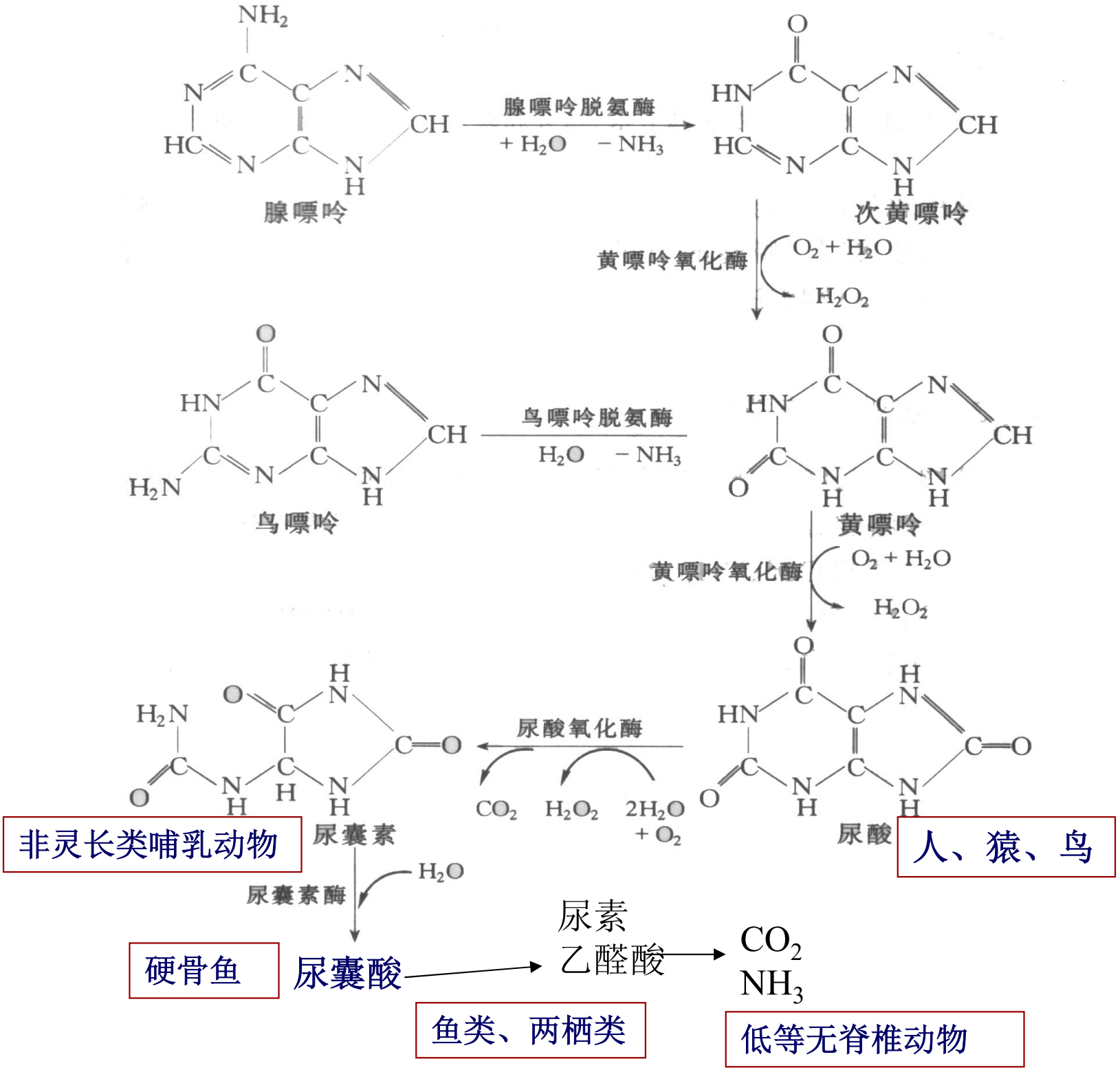
1、嘌呤的降解

2、嘧啶的降解

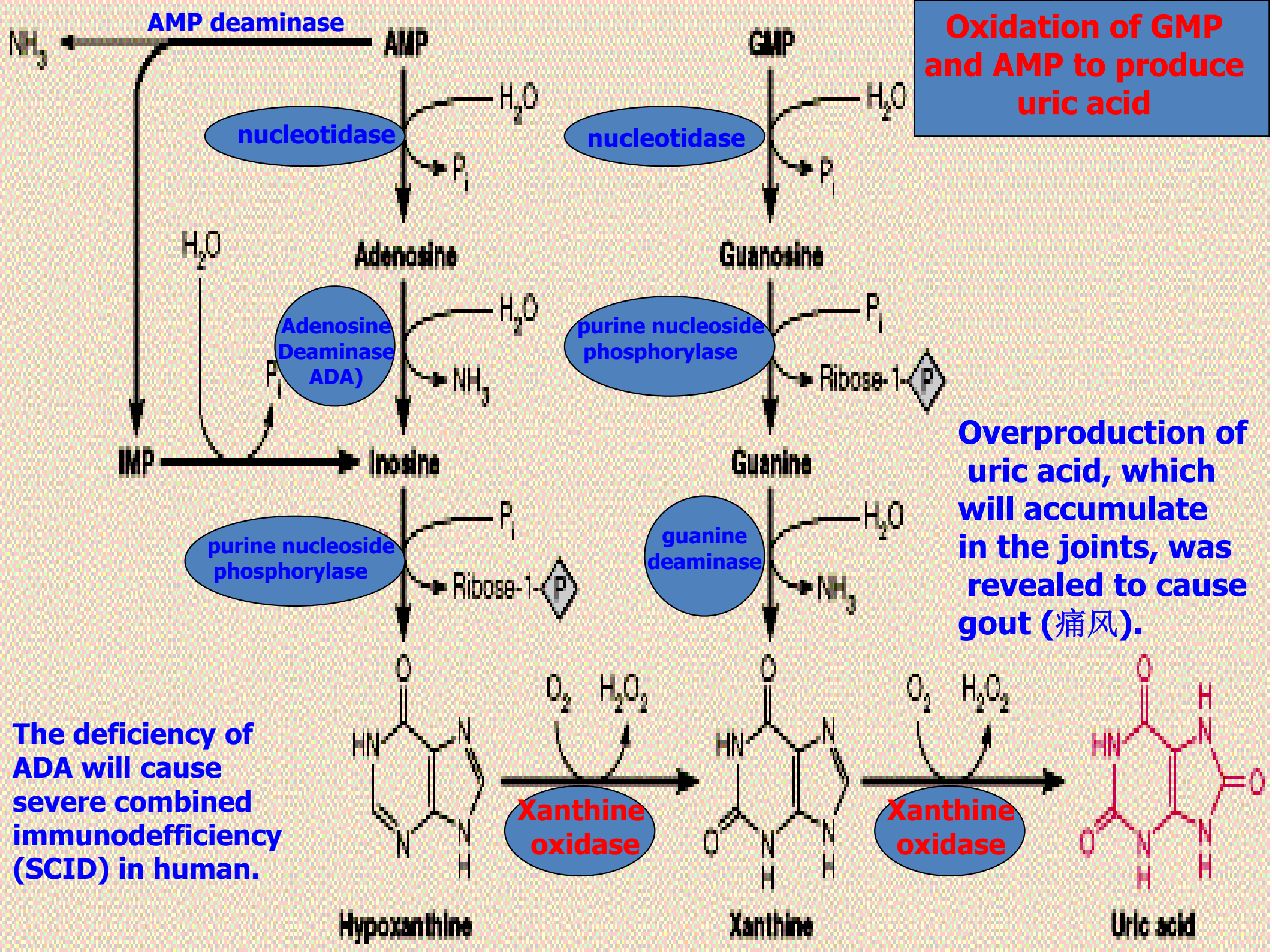
嘌呤的分解

- 腺嘌呤和鸟嘌呤要经过脱氨氧化转变为黄嘌呤再进行降解
- 不同生物分解嘌呤的酶系不一样
- 生物进化程度愈高，分解嘌呤的能力愈差




腺嘌呤
腺嘌呤脱氨酶
+ H2O − NH3
次黄嘌呤
黄嘌呤氧化酶
O2 + H2O
H2O2
鸟嘌呤
鸟嘌呤脱氨酶
H2O − NH3
黄嘌呤
黄嘌呤氧化酶
O2 + H2O
H2O2
尿酸氧化酶
CO2 H2O2 2H2O + O2
尿囊素
尿酸
非灵长类哺乳动物
人、猿、鸟
尿囊素酶
H2O
硬骨鱼
尿囊酸
尿素
乙醛酸
CO2
NH3
鱼类、两栖类
低等无脊椎动物

Oxidation of GMP and AMP to produce uric acid



Overproduction of uric acid, which will accumulate in the joints, was revealed to cause gout (痛风).

The deficiency of ADA will cause severe combined immunodefficiency (SCID) in human.

Uric acid is the excreted end product of purine nucleotide catabolism in humans and many other animals.



- 关键酶：黄嘌呤氧化酶
- 抑制剂—别嘌呤醇
- 最终产物：尿酸

正常人：血尿酸 119-357 μmol/L（2-6 mg/dl）

男 267.7 μmol/L（4.5 mg/dl）

女 208.2 μmol/L（3.5 mg/dl）

血尿酸 > 8 mg/dl

↓

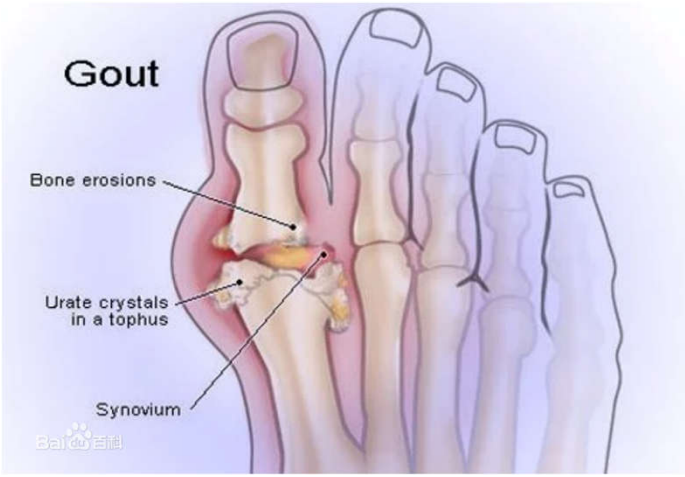
结晶

↓

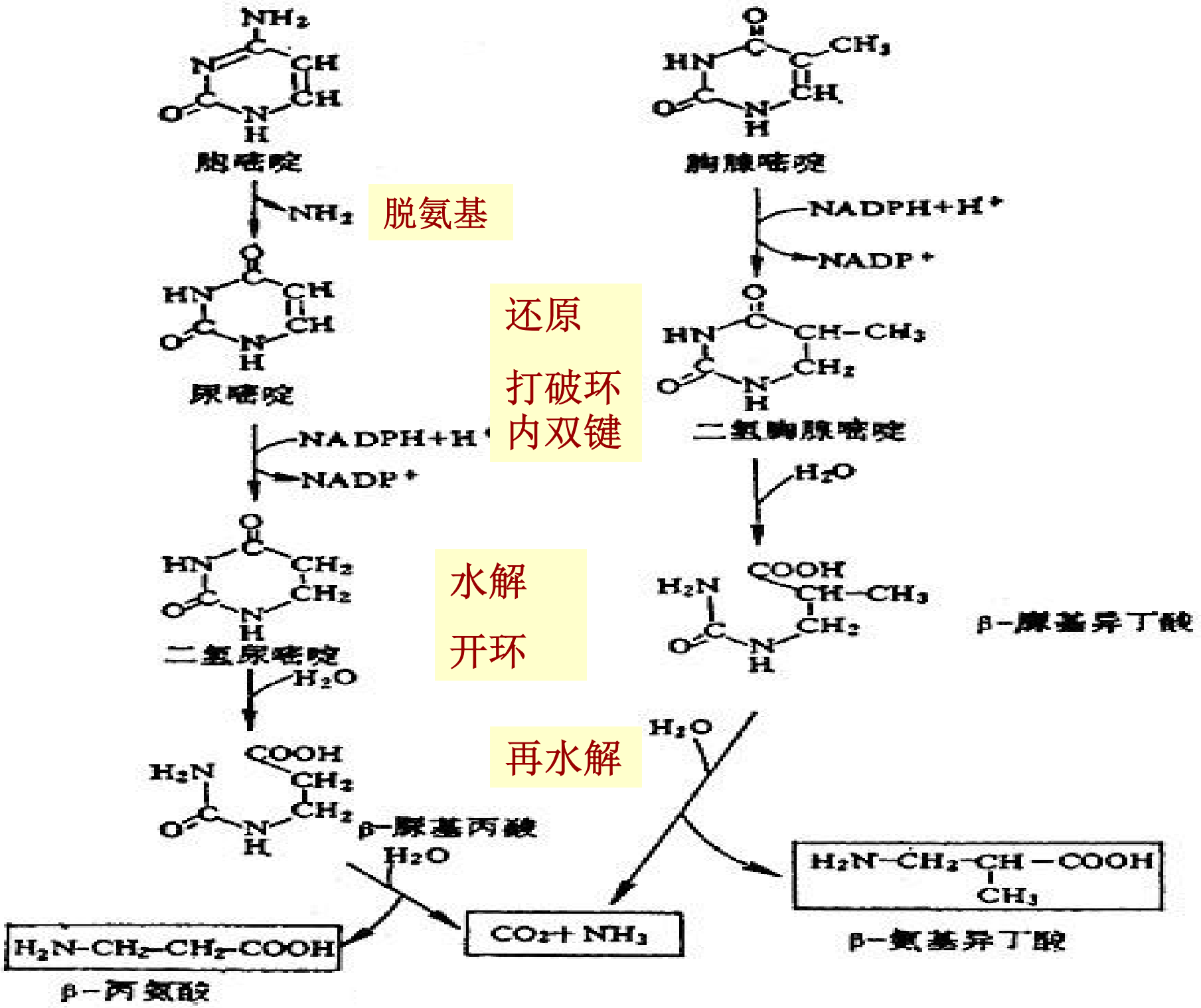
沉积组织

↓

痛风（gout）



嘧啶的分解


胞嘧啶
脱氨基
胸腺嘧啶
NADPH+H+
NADP+
尿嘧啶
还原
打破环内双键
二氢胸腺嘧啶
NADPH+H+
NADP+
H2O
二氢尿嘧啶
水解
开环
β-脲基异丁酸
H2O
β-脲基丙酸
再水解
H2O
H2N-CH2-CH2-COOH
β-丙氨酸
CO2+NH3
H2N-CH2-CH-COOH
CH3
β-氨基异丁酸

END

Thank you

